TASMIHA HASSAN

Software Engineer

Vancouver, BC | +1 (289) 654-7499 | htasmiha@gmail.com | linkedin.com/in/thas | tasmainian.github.io

PROFESSIONAL SUMMARY

Software engineer with 5+ years of experience building large-scale cloud security products at Microsoft. Full-stack across React/TypeScript, C#/.NET, and Scala/Spark on Azure, owning features from design through deployment and monitoring, and using AI tooling to turn manual workflows into reusable automation. Co-inventor on an AI agent risk patent.

EXPERIENCE

Microsoft · **Software Engineer → Software Engineer II** Jun 2021 – Oct 2026
Microsoft Purview Insider Risk Management — enterprise user and AI agent risk detection. Vancouver, Canada

- Resolved customer escalations **within SLA**, joining customer calls to investigate directly, and turned one enterprise customer's request into the design for alert event correlation
- Extended **20 REST endpoints** and Azure Event Hubs ingestion in **two months** to ship **AI agent risk detection**
- Shipped proactive onboarding cards in the Defender portal that walk admins to their first policy, contributing to **68% of opted-in tenants** reaching active-admin status with **7.8% month-over-month growth**
- Hardened an agent session scoring pipeline in **Scala/Spark**, writing the scoring logic, validation, record-type routing and time-gated state merging behind **2 new agent signals**
- Designed and presented a **failure-recovery architecture** for streaming jobs on a pipeline processing **~45 TB/day** of Microsoft 365 activity, generalized for **reuse by new jobs as the system scales**
- Migrated alerts, insights, and role-based access control into the Microsoft Defender portal and ran the release process on the team's Azure DevOps pipeline, achieving **36% product adoption** and **30x exposure** to potential tenants
- Built a **serverless** durable orchestrator on Azure Functions producing daily tenant-wide adoption metrics, feeding a **Kusto dashboard** reviewed in weekly service reviews
- Cut a multi-week manual deployment to **three days** by automating Event Hub, Key Vault, and managed identity provisioning across **56 Azure regions** in PowerShell, with tooling reusable for **government cloud** deployment
- Owned the Priority Content Only Scoring feature end-to-end, from React/TypeScript UI to C#/.NET scoring across **4 independent services**, reducing noisy alerts for enterprise customers by **15%**
- Wrote the post-mortem on production timeouts in Kusto queries joining **millions of records**, tracing them to full-history scans and an inefficient join strategy; a single pass with batched joins took **p99 from timeout to under 3 seconds**

McMaster University Hamilton, Canada
Program Support Assistant, Faculty of Engineering Jan 2021 – Apr 2021

- Took an accessibility-focused **React Native** and **Firebase** app for users with cognitive disabilities **from 0 → 1**

Embross Toronto, Canada
Software Engineer Intern May 2019 – Aug 2020

- Improved production stability across 2 airline workflows by **~15%**, resolving **55+ defects** and delivering on-time production patches in a **Java Spring MVC** application
- Traveled on site to gather requirements directly from airline stakeholders, then led full-stack implementation of a frequent flyer loyalty program enabling kiosk check-in benefits for **~3 million passengers**
- Designed and implemented full-stack tooling for airline check-in systems, automating configuration and PDF workflows in React, Node.js, Express, and SQL, cutting engineering configuration time from **15 minutes to 1 minute**

TECHNICAL SKILLS

Languages	C#, Java, JavaScript, TypeScript, Scala, Python, C, SQL, KQL
Frameworks	.NET, React, React Native, Node.js, Express, Redux, Spark, Spring MVC
Cloud & Data	Azure (Event Hubs, Functions, Key Vault, App Service), Cosmos DB, Kusto/ADX
Tools	REST APIs, CI/CD (Azure DevOps), Git, PowerShell, automated testing, observability
AI	OpenAI API, MCP, prompt engineering, AI workflows, Claude Code, GitHub Copilot

PATENT

Artificial Intelligence Agent Risk Assessment and Mitigation Filed with Microsoft, 2025
Co-inventor of a system for detecting risky AI agent behavior in enterprise environments and enforcing policy in real time

PROJECTS

Rackwise · AI coach for the NYT Crossplay word game Go, Python, Postgres, Docker
MCP server for word lookup, **deterministic rules and scoring**, and **RAG** with local embeddings for the coaching prompt.

EDUCATION

McMaster University Hamilton, Canada
Bachelor of Engineering, Software Engineering Sep 2016 – Apr 2021
Relevant Coursework: Large Systems Design, Data Structures and Algorithms, Software Testing, Databases, ML